

SafeStrain™

Improved Pasta Strainer

By: Aleia, Alexa, Amaris, Arth, Irving

Product and User Description



Common Kitchen tool used by wide range of users



Task requires handling boiling water, heavy pots, and awkward wrist positions



High injury risk due to heat, weight, and lack of stability



Redesign must account for physical, perceptual, and safety needs of diverse users

Market & Use Context

Universally used: performance directly affects safety and satisfaction



Pain Points Identified:

Two-Handed coordination
required (Pot + Strainer)

Steam obstructs visibility and
causes uncomfortability

Heavy loads and lack
of malleability cause wrist strain

High likelihood of spills/burns



Redesign goal: improve task flow, reduce effort, enhance control

Issues



Pasta easily falls out of the strainer



Heat/steam poses a threat to the use



Small, slippery handles, reduced grip force



Wrist strain from the angle required



Solutions



Increase the surface area of the strainer



Add steam-direction vents + a small shield



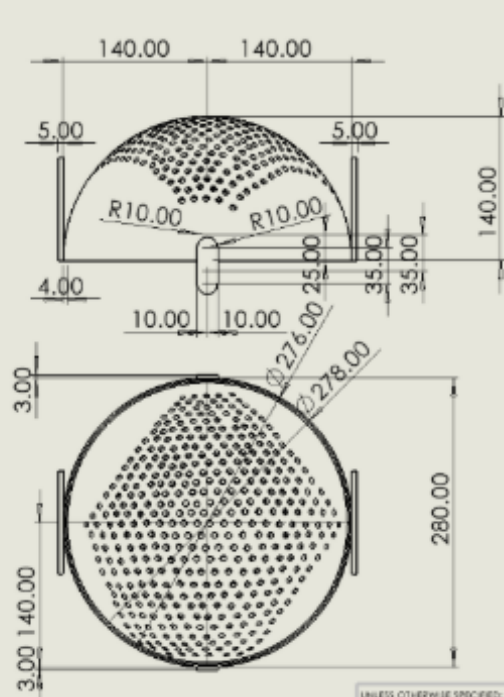
Tactile feedback, a click when clip locks



Use a 30-35 mm textured, slightly soft handle to reduce grip force

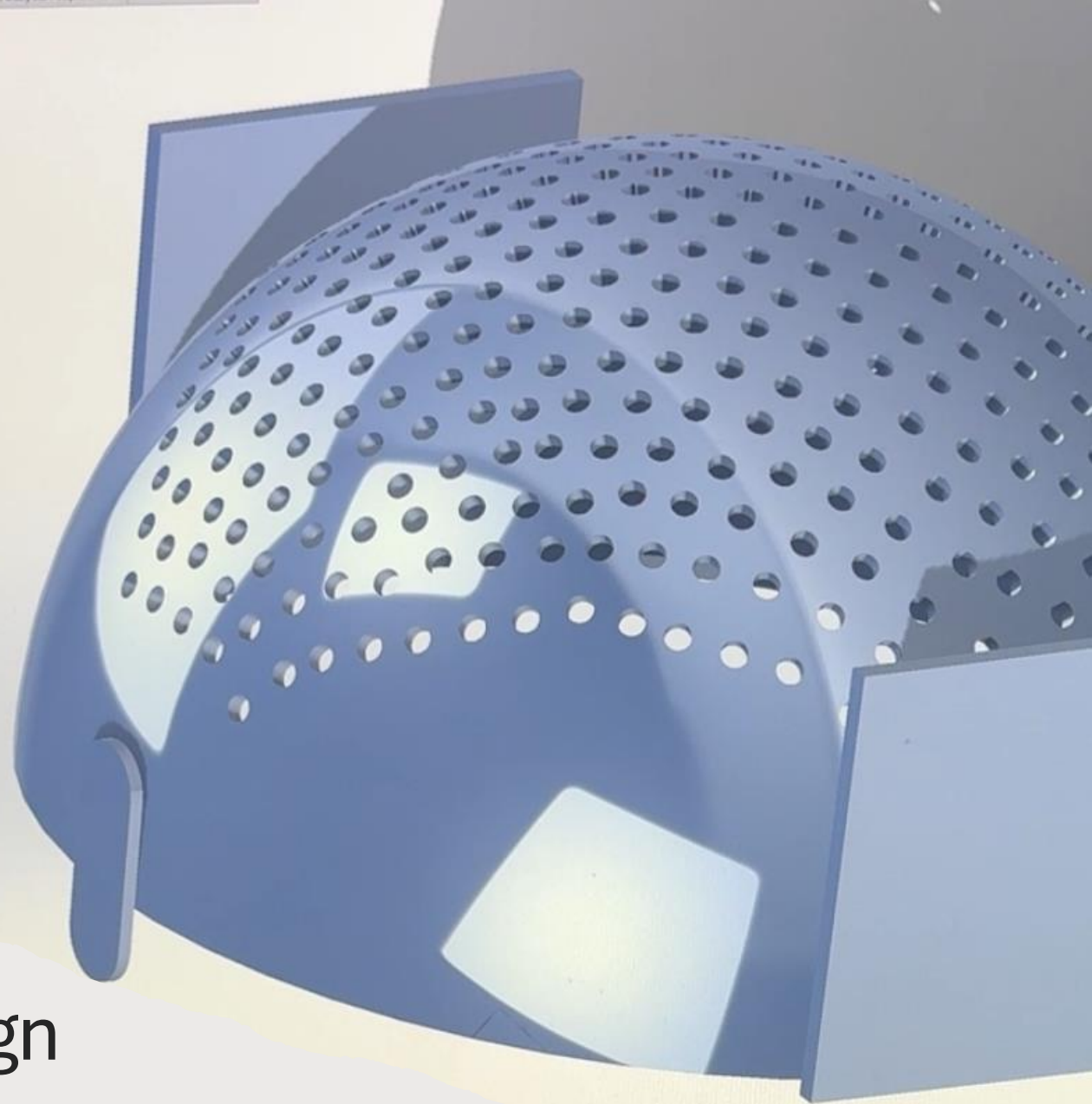


Insulated, heat-resistant materials on handles



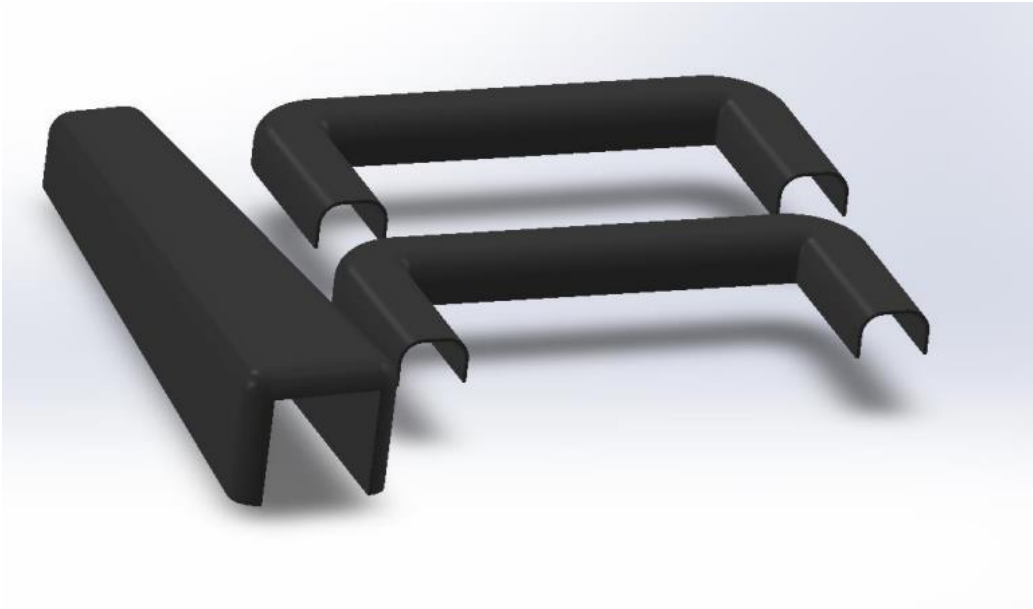
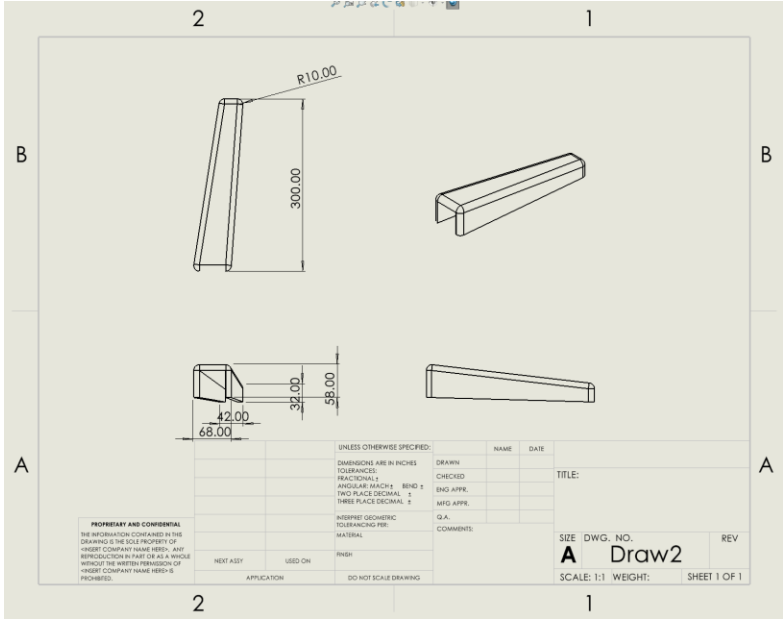
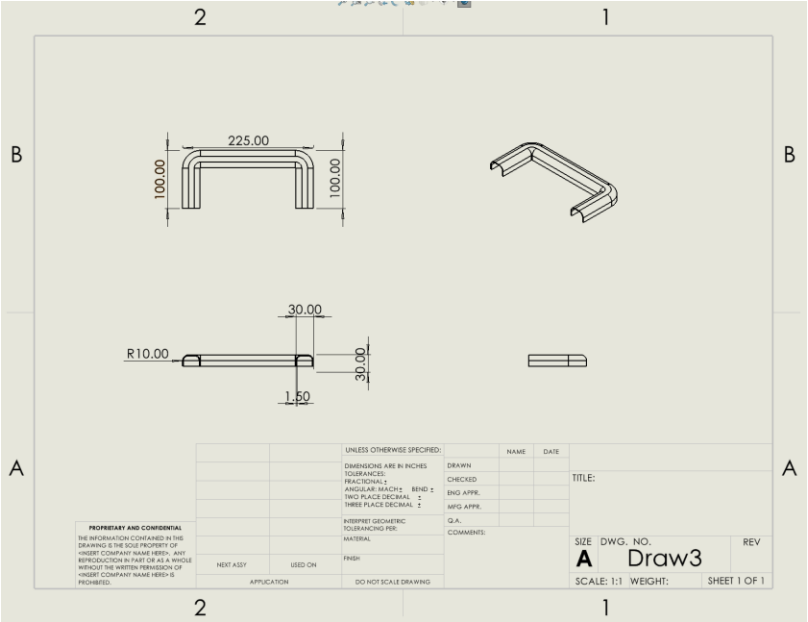
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DESIGNER AND CHECK SHARP EDGES		DO NOT SCALE DRAWING	
NAME	SIGNATURE	DATE				TITLE:	
DESIGN							
CHECKED							
APPROVED							
ENG							
Q.A.							
				MATERIAL:		DWG NO.	
				WEIGHT:		SCALE 1:1	
						SHEET	

Strainer



Prototype- Solid Works / Design

Prototype-Solid Works Design



Manufacturability & Cost Considerations

Manufacturability:

- Using materials like ; Polypropylene (basic plastic polymer) injection molding allows the entire strainer (bowl, rim, holes, handle) to be made in one shot, reducing steps.
- Staggered circular holes and no undercuts mean basic two-plate tooling can be used , hence cheaper and faster to produce.
- If the handle is separate, a snap-fit connection allows quick, tool-free assembly.
- Low material usage (120–200 g PP) keeps production cost and cycle time low.

Cost overview :

- PP(Polypropylene) is cheap and widely available (~\$1.50–\$2.00/kg), keeping raw material costs low.
- Circular holes are easy to form, small tooling cost (~\$0.05 per strainer).
- Lightweight design (~150 g per strainer) → ~\$0.25 in material which implies lower shipping cost.
- Simple single-piece design, no coatings or extra parts which in turn reduces assembly & finishing cost.
- Estimated total production cost per strainer: ~\$0.50–\$0.60, making it affordable for customers.

Potential impacts



Increased User Safety: Lower burn risk + lower chances of dropping pot



Improved Physical Comfort: Reduced wrist strain + wider range of hand sizes



Greater Usability & Error Reduction: Fewer mistakes + More intuitive task flow for inexperienced users



Broader User Accessibility: Older adults, students and low-strength users + diverse perceptual and motor abilities

Conclusion

- Our prototype addresses key human-factor issues by:
 - Reduces user errors
 - Increases safety
 - Lowers physical strain
- We envision this product becoming a standard in households, serving as a safer, more effective alternative to traditional strainers.